

D340

Cyber Defense and Countermeasures

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

Analysts at a county government confirm that malware is spreading from a finance workstation to other hosts. The county IT budget was cut by 10% this year. The workstation belongs to a finance clerk who has worked there for 22 years. What should the team do first?

- A. Write the lessons-learned report.
- B. Interview the clerk about her long tenure.
- C. Reimage the workstation right away.
- D. Isolate the affected hosts from the network to stop the spread.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A ransomware incident has been contained, the malware removed, and systems restored. Which activity comes next?

- A. Detection and analysis of the same incident.
- B. Nothing; once restored, the incident is fully closed.
- C. A lessons-learned review to improve controls and the response plan.
- D. Containment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A compromised server at a law firm is still running. The firm expects the case may go to court. The server was installed by a contractor in 2021. What should the responder collect first?

- A. Copy the disk image first, since disks hold the most data.
- B. The contractor's installation records.
- C. A capture of system memory, recorded with chain-of-custody documentation.
- D. Power off the server, then image the disk.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A database server is valued at \$200,000. A certain outage would destroy 25% of its value, and it is expected once every two years. The server's vendor won an industry award last year. What is the annualized loss expectancy?

- A. \$200,000
- B. \$25,000
- C. \$50,000
- D. \$100,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A risk has an ALE of \$30,000 and an SLE of \$120,000. What is the annualized rate of occurrence?

- A. 0.25 (once every four years)
- B. 1
- C. 4
- D. \$90,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A factory's control workstation runs software that cannot be patched for six months without halting production. A critical flaw in it is being exploited in the wild. The plant manager has twenty years of experience. What is the best action now?

- A. Patch today and halt production regardless of cost.
- B. Apply compensating controls such as isolating the workstation and restricting access, and track the risk until it can be patched.
- C. Ask the plant manager to accept all risk because of his experience.
- D. Do nothing until the patch window.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

The board of a regional bank, which meets quarterly in its downtown headquarters, wants to know which threats to the financial sector are rising over the next year to guide budget decisions. What kind of threat intelligence should the security team give them?

- A. Operational intelligence about one specific campaign's steps.
- B. Raw firewall logs.
- C. Tactical intelligence, such as a list of malicious IP addresses.
- D. Strategic intelligence.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A SIEM rule fires on an admin logging in at 2 a.m. The analyst confirms it was a scheduled maintenance task approved last week. The SOC recently moved to a new ticketing tool. How should the alert be classified?

- A. False positive.
- B. True positive.
- C. False negative.
- D. A ticketing tool error.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

After an attacker moved data out over several weeks, the review finds no alert ever fired. What does this represent, and what is the risk of the fix?

- A. A true negative; no action needed.
- B. A false positive; the rule should be disabled.
- C. A false negative; making rules more sensitive may raise false positives.
- D. A false negative; tuning up has no downside.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A clinic installs a system that records every access to patient files and flags unusual access patterns for review. The system was chosen after a six-month vendor search. What function does this control serve?

- A. Physical.
- B. Preventive.
- C. Corrective.
- D. Detective.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which is an example of a corrective control?

- A. Restoring files from backup after ransomware.
- B. Security awareness training.
- C. A firewall rule blocking a port.
- D. An intrusion detection alert.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

During an incident, a university confirms that student records were exposed. A reporter calls the help desk, and a technician who has worked there for two semesters wants to explain what happened. What should happen?

- A. Let the technician explain the details to show transparency.
- B. Refer the reporter to the designated spokesperson, and let legal and leadership handle notifications required by law.
- C. Say nothing to anyone, including affected students.
- D. Wait until the technician has more experience.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

Analysts at a county government confirm that malware is spreading from a finance workstation to other hosts. ~~The county IT budget was cut by 10% this year.~~ The workstation belongs to a finance clerk ~~who has worked there for 22 years.~~ What should the team do first?

Target word: next phase **Decides:** Active spread means containment comes first.

A	Confusion set	Post-incident activity comes after recovery.
B	Camouflage	Her tenure does not decide the response step.
C	Inverted operation	Eradicating first loses evidence and misses other infected hosts.
D	Correct	Containment limits damage before eradication and recovery.

2. Answer C Concept 1 • transfer

A ransomware incident has been contained, the malware removed, and systems restored. Which activity comes next?

Target word: phase after recovery **Decides:** Recovery is done; post-incident activity follows.

A	Inverted operation	That phase came earlier.
B	Absolute language	Skipping review repeats the same failures.
C	Correct	Post-incident activity closes the lifecycle.
D	Confusion set	Already completed.

3. Answer C Concept 2

A compromised server at a law firm is still running. The firm expects the case may go to court. ~~The server was installed by a contractor in 2021.~~ What should the responder collect first?

Target word: first evidence to collect **Decides:** Memory is the most volatile source.

A	Dropped element	Drops the volatility order.
B	Camouflage	Useful context, but not volatile evidence.
C	Correct	RAM is lost at shutdown; documentation keeps evidence admissible.
D	Inverted operation	Shutting down destroys memory evidence.

4. Answer B Concept 3

A database server is valued at \$200,000. A certain outage would destroy 25% of its value, and it is expected once every two years. ~~The server's vendor won an industry award last year.~~ What is the annualized loss expectancy?

Target word: ALE **Decides:** $SLE = \$200,000 \times 0.25 = \$50,000$; $ARO = 0.5$.

A	Wrong stem number	The asset value, not the loss expectancy.
B	Correct	$\$50,000 \times 0.5 = \$25,000$.
C	Dropped element	That is SLE; the frequency is left out.
D	Inverted operation	Multiplies by 2 instead of 0.5.

5. Answer A Concept 3 · transfer

A risk has an ALE of \$30,000 and an SLE of \$120,000. What is the annualized rate of occurrence?

Target word: solve for ARO **Decides:** $ARO = ALE / SLE$.

A	Correct	$\$30,000 / \$120,000 = 0.25$.
B	Absolute language	Assumes every risk happens once a year.
C	Inverted operation	Divides SLE by ALE.
D	Adjacent category	Subtracts instead of dividing; ARO is a rate, not a dollar amount.

6. Answer B Concept 4

A factory's control workstation runs software that cannot be patched for six months without halting production. A critical flaw in it is being exploited in the wild. ~~The plant manager has twenty years of experience.~~ What is the best action now?

Target word: response when patching is not possible **Decides:** No patch possible soon, and the flaw is being exploited.

A	Absolute language	Ignores business impact when a compensating option exists.
B	Correct	Reduces exposure while the permanent fix waits.
C	Camouflage	Experience does not decide risk treatment.
D	Dropped element	Drops the active exploitation that raises urgency.

7. Answer D Concept 5

The board of a regional bank, ~~which meets quarterly in its downtown headquarters,~~ wants to know which threats to the financial sector are rising over the next year to guide budget decisions. What kind of threat intelligence should the security team give them?

Target word: intel level for the board **Decides:** Leaders making long-term budget decisions.

A	Adjacent category	Detailed campaign methods serve defenders planning responses.
B	Confusion set	Data, not intelligence.
C	Inverted operation	Indicators serve tools and analysts, not the board.
D	Correct	High-level trends for leadership decisions.

8. Answer A Concept 6

A SIEM rule fires on an admin logging in at 2 a.m. The analyst confirms it was a scheduled maintenance task approved last week. ~~The SOC recently moved to a new ticketing tool.~~ How should the alert be classified?

Target word: alert classification **Decides:** Alert fired; the activity was benign.

A	Correct	Flagged, but not malicious.
B	Inverted operation	That would mean real malicious activity.
C	Confusion set	A false negative is a missed real event.
D	Camouflage	The tool change did not cause the alert.

9. Answer C Concept 6 · transfer

After an attacker moved data out over several weeks, the review finds no alert ever fired. What does this represent, and what is the risk of the fix?

Target word: classify the miss and tuning trade-off **Decides:** A real event with no alert.

A	Confusion set	A real attack happened.
B	Inverted operation	Nothing fired, so it cannot be a false positive.
C	Correct	Tuning trades missed detections against noise.
D	Absolute language	More sensitivity usually means more noise.

10. Answer D Concept 7

A clinic installs a system that records every access to patient files and flags unusual access patterns for review. ~~The system was chosen after a six-month vendor search.~~ What function does this control serve?

Target word: control function **Decides:** Records and flags after access happens.

A	Adjacent category	A control type, not a function, and this one is technical.
B	Inverted operation	It does not block access.
C	Confusion set	It does not restore anything.
D	Correct	Finds and flags events; it does not stop them.

11. Answer A Concept 7 · transfer

Which is an example of a corrective control?

Target word: pick the corrective control **Decides:** Corrective controls restore after an event.

A	Correct	Returns systems to normal after damage.
B	Adjacent category	Administrative and preventive.
C	Inverted operation	Preventive.
D	Confusion set	Detective.

12. Answer B Concept 8

During an incident, a university confirms that student records were exposed. A reporter calls the help desk, and a technician ~~who has worked there for two semesters~~ wants to explain what happened. What should happen?

Target word: communication handling **Decides:** Personal data exposure triggers legal duties and a set communication plan.

A	Inverted operation	Unapproved statements can break legal and investigation needs.
B	Correct	The plan controls external statements and legal notice duties.
C	Absolute language	Notification laws may require telling affected people.
D	Camouflage	Experience is not the issue; role and plan are.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	4	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	11	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Incident response lifecycle	Preparation → detection and analysis → containment, eradication, and recovery → post-incident activity; contain before you eradicate.	Wiping and rebuilding before the threat is contained and scoped.	An active incident and a choice of what to do first.
2	Order of volatility and chain of custody	Collect the most volatile evidence first (memory before disk) and document every hand-off.	Powering off a system first, losing memory contents.	A live, compromised machine and a need to preserve evidence.
3	Quantitative risk (SLE, ARO, ALE)	$SLE = \text{asset value} \times \text{exposure factor}$; $ALE = SLE \times ARO$.	Using asset value as SLE, or adding instead of multiplying.	An asset value, a percentage loss, and a yearly frequency.
4	Vulnerability prioritization and compensating controls	Prioritize by severity, asset criticality, exposure, and known exploitation; when a patch is impossible, apply a compensating control.	Leaving an unpatchable critical system as is because a patch is not available.	A system that cannot be patched soon.
5	Threat intelligence levels	Strategic intel informs leaders on trends; operational intel describes campaigns and actor methods; tactical intel gives indicators such as hashes and IP addresses for tools.	Handing executives raw indicator lists as 'strategy'.	Who will use the intel and for what decision.
6	Alert outcomes and tuning	True positive = real event flagged; false positive = benign event flagged; false negative = real event missed; tuning balances alert noise against missed detections.	Tuning away every noisy rule and creating blind spots.	An alert that turned out benign, or an attack with no alert.
7	Control functions	Preventive controls stop events; detective controls find them; corrective controls restore after them; compensating controls substitute when the primary control is not possible.	Calling logging or cameras preventive.	Whether the control acts before, during, or after an event.
8	Communication and legal duties during incidents	Follow the communication plan: involve legal and leadership, meet notification laws and contracts, and avoid unapproved public statements.	Staff posting details or notifying outsiders on their own.	Personal data exposed and pressure to say something fast.

Course scope (public description): Traditional defenses—such as firewalls, security protocols, and encryption—sometimes fail to stop attackers determined to access and compromise data. This course provides the fundamental skills to handle and respond to computer security incidents in an information system. The course addresses various underlying principles and techniques for detecting and responding to current and emerging computer security threats. Students learn how to leverage intelligence and threat detection techniques; analyze and interpret data; identify and address vulnerabilities; suggest preventative measures; effectively respond to and recover from incidents; and handle various types of incidents, risk assessment methodologies, and various laws and policies related to incident handling. This course prepares students for the CompTIA Cybersecurity Analyst

(CySA+) certification exam. The following courses are prerequisites: Networks and Network and Security - Applications.